MANIKANDAN S

Design Verification Engineer
+91 9751263491 | dhanushmani379@gmail.com | LinkedIn | GitHub

Professional Summary

Design Verification Engineer with hands-on experience in RTL Design and Functional Verification using Verilog and SystemVerilog. Skilled in AMBA protocols (APB, AHB, AXI), UVM, SVA, STA and EDA tools including QuestaSim, VCS, Vivado, and Design Compiler. Strong in problem-solving, collaboration, and continuous learning, with a solid foundation in Digital Design and ASIC Front-End Verification.

Technical Skills

- **Digital Design:** Combinational & Sequential Circuits, FSMs, Memories, CMOS Fundamentals.
- **HDL/Hardware:** Verilog, SystemVerilog (OOP, Constraints, Assertions, Functional Coverage).
- **RTL Design:** Synthesizable Verilog, RTL Coding, Static Timing Analysis (STA), Code Coverage.
- **Verification:** UVM, Layered Testbench Architecture, Scoreboard, Functional Coverage, SVA, Regression testing.
- **Protocols:** APB, AHB, AXI, UART, SPI, I2C.
- **ASIC/FPGA:** ASIC Front-End Design Flow, FPGA Design Flow, Synthesis.
- **Processor Architecture:** RISC-V.
- **EDA Tools:** QuestaSim, Synopsys VCS, Synopsys Design Compiler, Xilinx Vivado, Xilinx ISE, EDA Playground.
- **Programming & Scripting:** C, Python, Tcl, Perl, Linux Commands.

Professional Training

AI Driven VLSI Design and Verification — VLSIGURU Oct 2026 – Present

- Designed synthesizable RTL modules and performed functional verification using SystemVerilog and UVM in industry-standard simulation environments (QuestaSim, VCS).
- Gained expertise in UVM Testbench Architecture, constrained-random stimulus generation, functional coverage, scoreboard-based checking, SVA and STA.

Projects

AHB VIP — UVM Verification **May 2026 – Jun 2026**

- Developed UVM-based verification environments for AHB VIP using constrained-random testing, scoreboards, and SystemVerilog Assertions (SVA).
- Achieved 100% functional coverage and verified protocol compliance by debugging transaction mismatches through coverage-driven verification and QuestaSim waveform analysis.

APB-based UART IP Core — RTL Design & UVM Verification **Dec 2026 – Jan 2027**

- Designed a 16550 UART IP with APB interface, programmable baud rate, interrupts, FIFO buffering, and loopback support.
- Verified the design using a UVM environment with RAL, 100% functional coverage, and scoreboard-based end-to-end validation.

HydroPi – Smart Hydroponic Plant Growth and Disease Monitoring System | *IoT & Deep Learning*

- Developed an IoT-based hydroponic monitoring system using Raspberry Pi and Arduino to collect real-time pH, temperature, humidity, and water-level data with Firebase cloud integration.
- Implemented a MobileNetV3-based AI model for disease detection in Tomato, Bell Pepper, and Grapevine plants, with a Flutter app for live monitoring and alerts.

Education

Anna University Regional Campus, Coimbatore 2022-2026
Bachelor of Engineering in Electronics and Communication Engineering
*(**Honors** with VLSI Specialization)* *(CGPA: 8.3 / 10.00)*

Little Flower Matric Higher Secondary School April 2022
HSC: 89.33 % *Perambalur, Tamil Nadu*

Little Flower Matric Higher Secondary School May 2020
SSLC: 95 % *Perambalur, Tamil Nadu*

Learning Certification